

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. Which software engineering principle recommends solving a complex problem by dividing it into smaller independent modules?

- A. Coupling
- B. Decomposition
- C. Encapsulation
- D. Generalization

Answer: B

2. A project manager decides to reuse an existing authentication library instead of developing one from scratch. Which software engineering principle is being applied?

- A. Refactoring
- B. Software Reuse
- C. Requirement Traceability
- D. Verification

Answer: B

3. Which of the following activities is performed BEFORE software coding begins?

- A. Integration Testing
- B. Architectural Design
- C. Deployment
- D. Regression Testing

Answer: B

4. Which document provides detailed instructions for installing and configuring software?

- A. Test Plan
- B. Installation Guide
- C. Requirement Specification
- D. Design Document

Answer: B

5. Which software quality attribute ensures that unauthorized users cannot access sensitive information?

- A. Reliability
- B. Maintainability
- C. Security
- D. Portability

Answer: C

6. Which software engineering activity evaluates whether the project is technically and economically practical?
- A. Risk Management
 - B. Feasibility Study
 - C. Software Deployment
 - D. Code Inspection

Answer: B

7. Which estimation technique is commonly used when historical project data are available?
- A. Expert Judgment
 - B. Analogy-Based Estimation
 - C. Random Estimation
 - D. Brainstorming

Answer: B

8. Which software testing technique focuses on checking the internal logic of the source code?
- A. Black-box Testing
 - B. White-box Testing
 - C. Acceptance Testing
 - D. Beta Testing

Answer: B

9. Which software engineering practice attempts to identify defects without executing the software?
- A. Code Review
 - B. Regression Testing
 - C. Load Testing
 - D. Smoke Testing

Answer: A

10. Which UML diagram primarily illustrates the static structure of a software system?
- A. Activity Diagram
 - B. Sequence Diagram
 - C. Class Diagram
 - D. State Diagram

Answer: C

11. Which project management constraint is MOST directly affected when additional software features are introduced?
- A. Scope
 - B. Temperature
 - C. Processor Speed
 - D. Programming Language

Answer: A

12. Which software metric measures the average number of defects per thousand lines of source code?
- A. Function Point

- B. Defect Density
- C. Cyclomatic Complexity
- D. Code Coverage

Answer: B

13. Which software configuration management activity records every approved modification made to project artifacts?

- A. Version Control
- B. Refactoring
- C. Deployment
- D. Validation

Answer: A

14. Which software engineering objective focuses on minimizing the impact of future software modifications?

- A. Portability
- B. Maintainability
- C. Availability
- D. Performance

Answer: B

15. Which software review technique requires a formal meeting with assigned reviewer roles?

- A. Walkthrough
- B. Inspection
- C. Pair Programming
- D. Demonstration

Answer: B

16. Which risk management activity determines the probability and impact of identified risks?

- A. Risk Analysis
- B. Risk Avoidance
- C. Risk Monitoring
- D. Risk Reporting

Answer: A

17. Which software engineering principle promotes hiding implementation details from users?

- A. Encapsulation
- B. Cohesion
- C. Coupling
- D. Abstraction

Answer: D

18. Which software testing level verifies that the complete system satisfies specified requirements?

- A. Unit Testing
- B. Integration Testing
- C. System Testing

D. Mutation Testing

Answer: C

19. Which project document establishes the official approval to begin software development?

- A. Project Charter
- B. User Manual
- C. Test Report
- D. Maintenance Log

Answer: A

20. Which software engineering activity confirms that the developed software satisfies customer expectations?

- A. Verification
- B. Validation
- C. Refactoring
- D. Configuration Management

Answer: B